

Experiment No: 11

Date: 17/09/2026

Title: Mobile Number Validation using Exception Handling

Objective / Aim

To design and implement a Mobile Number Validation program in Java using exception handling and a user-defined exception, demonstrating concepts such as try, catch, throw, and custom exceptions.

System Requirements

Hardware

- Standard PC/Laptop
- Intel Core i3/i5 processor or equivalent
- Minimum 4 GB RAM
- Minimum 500 MB free storage

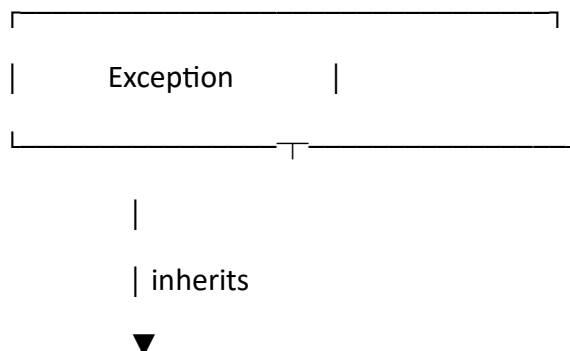
Software

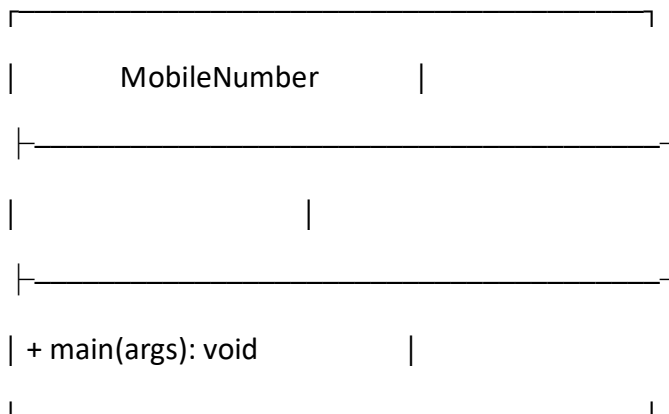
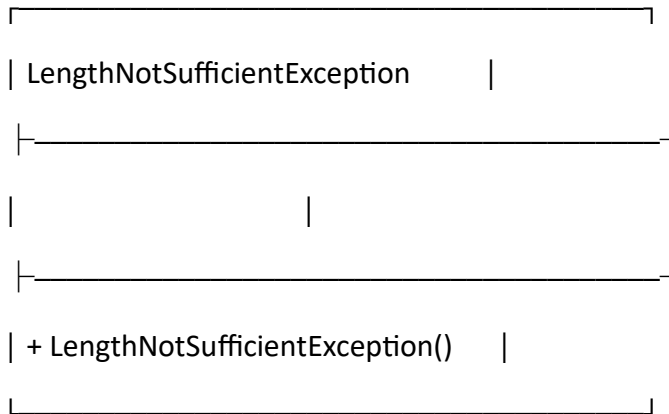
- Operating System: Windows / Linux / macOS
- Java Development Kit (JDK 17 or JDK 21)
- Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code
- Command Prompt / Terminal

Algorithm / Logic Description

1. Create a user-defined exception class named LengthNotSufficientException.
2. Extend the class from the built-in "Exception" class.
3. Define a constructor with an appropriate error message.
4. Create the main class named MobileNumber.
5. Create a "Scanner" object to accept the mobile number from the user.
6. Read the mobile number as a "String".
7. Place the validation logic inside a "try" block.
8. Check whether the mobile number contains more than 10 digits.
9. If the length is greater than 10, throw an "ArrayIndexOutOfBoundsException".
10. Check whether the mobile number contains fewer than 10 digits.
11. If the length is less than 10, throw the user-defined "LengthNotSufficientException".
12. Check each character of the mobile number using "Character.isDigit()".
13. If any character is not a digit, throw a "NumberFormatException".
14. Handle each exception using separate "catch" blocks.
15. If all conditions are satisfied, display "Valid number".

UML Class Diagram





Relationship: "LengthNotSufficientException" inherits from the built-in "Exception" class. The "MobileNumber" class uses this user-defined exception for mobile number validation.

Source Code

```
package javacore;
```

```
import java.util.*;
```

```
// User-defined exception
```

```
class LengthNotSufficientException extends Exception {

    // Constructor with error message
    LengthNotSufficientException() {
        super("Invalid Mobile Number – LengthNotSufficientException");
    }
}

// Main class
public class MobileNumber {

    public static void main(String args[]) {

        // Create Scanner object
        Scanner sc = new Scanner(System.in);

        // Get mobile number from user
        System.out.print("Enter Mobile Number: ");
        String number = sc.next();

        try {

            // Check for more than 10 digits
            if (number.length() > 10) {
                throw new ArrayIndexOutOfBoundsException();
            }

            // Check for less than 10 digits
```

```
    if (number.length() < 10) {  
        throw new LengthNotSufficientException();  
    }  
  
    // Check each character  
    for (int i = 0; i < 10; i++) {  
  
        // If character is not a digit  
        if (!Character.isDigit(number.charAt(i))) {  
            throw new NumberFormatException();  
        }  
    }  
  
    // Mobile number is valid  
    System.out.println("Valid number");  
}  
  
// Catch excess length exception  
catch (ArrayIndexOutOfBoundsException e) {  
    System.out.println(  
        "Invalid Mobile Number-ArrayIndexOutOfBoundsException"  
    );  
}  
  
// Catch insufficient length exception  
catch (LengthNotSufficientException e) {  
    System.out.println(e.getMessage());  
}
```

```
// Catch non-digit exception
catch (NumberFormatException e) {
    System.out.println(
        "Invalid Mobile Number – NumberFormatException"
    );
}
}
}

---
```

Compilation & Execution Commands

Save the source code as:

MobileNumber.java

If you are using the "package javacore;" statement, save the file inside the appropriate "javacore" package/folder.

Compile the program:

```
javac MobileNumber.java
```

Run the program:

```
java MobileNumber
```

Output Screen

Example 1 – Valid Mobile Number

Enter Mobile Number: 9876543210

Valid number

Example 2 – More Than 10 Digits

Enter Mobile Number: 98765432101

Invalid Mobile Number-ArrayIndexOutOfBoundsException Exception

Example 3 – Less Than 10 Digits

Enter Mobile Number: 987654321

Invalid Mobile Number – LengthNotSufficientException

Example 4 – Contains Non-Digit Character

Enter Mobile Number: 98765A3210

Invalid Mobile Number – NumberFormatException

Result

The Mobile Number Validation System was successfully implemented in Java using exception handling and a user-defined exception. The program validates the mobile number based on its length and character type and handles invalid inputs using appropriate exception handling mechanisms.